FastSpeech 2 Text-to-Speech for English and Telugu

Technical report · Implementation of FastSpeech 2-based text-to-speech synthesis · 7 November 2026

Summary

Two FastSpeech 2 models were trained separately: English on AI4Bharat Svarah and Telugu on IndicTTS Telugu. Each used about 5 h of training audio and 1 h of validation audio, with no recording in both sets. Both start from the public `ming024/FastSpeech2` implementation of FastSpeech 2 (Ren et al., ICLR 2021), with its pretrained HiFi-GAN vocoder.

	English	**Telugu**
Dataset	AI4Bharat Svarah: Indian-accented English, 117 documented speakers	IndicTTS Telugu: female voice
Training / validation audio selected	4.9995 h / 1.0012 h	5.0044 h / 1.0018 h
Phonemes	ARPAbet: lexicon, g2p, heteronyms, numbers; variants chosen from the training alignments	New rule-based Telugu phonemizer, 46 phonemes
Forced aligner	Montreal Forced Aligner, pretrained `english_us_arpa`	Montreal Forced Aligner, trained on the selected Telugu data
Voice conditioning	109 demographic groups (a proxy, not verified speakers)	None (single voice)
Training	20,000 steps, 1.28 h	25,000 steps, 1.77 h
Final validation loss (total)	2.269	1.792
Inference voice	`Male-30-45-Maithili-Bihar-Darbhanga`, chosen by speech recognition; pitch range ×1.3	The single voice
ASR word error rate, synthesized validation speech (real recordings)	20.3% (7.2%)	Not measurable: the recognizer also fails on real Telugu recordings
Checkpoint	`English/checkpoints/model_only_20000.pth.tar`	`Telugu/checkpoints/model_only_25000.pth.tar`

Main findings

- **Telugu** was easy to understand on the first training run, including the loanword, the homograph పట్టు and the spoken date and time.
- **English** needed three training runs and two inference fixes. The final model trains on the cleanest Svarah recordings, conditioned on demographic groups. At inference it uses the pronunciations the aligner found most often in training, the most intelligible voice, and a wider pitch range. These changes cut the speech recognizer's word error rate on the English test sentences from 48% to 17.2%.
- **The difference between the languages comes mostly from the data.** Svarah has many voices recorded in varied, often noisy conditions; IndicTTS has one consistent studio voice. As a result:
  - the Telugu pitch predictor beats a text-free guess by 59%, against 6% for English;
  - the vocoder alone already loses far more on English recordings than on Telugu (Section 4.2).

About the evidence. Every number comes from logged values, generated files or the repository's scripts. Listening judgements are the author's own: there was no test with several listeners and no MOS score. Two objective measures stand in for a larger listening test: speech recognition with Whisper, and mel-cepstral

distortion.

1. Implementation approach

1.1 Architecture components used from the reference implementation

The model, losses and vocoder are the reference implementation's, without architectural changes:

Component	Configuration in both models
Phoneme embedding and encoder	4 feed-forward Transformer blocks, 256 dimensions, 2 attention heads, 1D convolutions (kernel sizes 9 and 1, 1024 filters)
Variance adaptor	Duration, pitch and energy predictors (2 convolutional layers each, 256 filters, dropout 0.5). Duration is predicted in the log domain, and the length regulator repeats each phoneme's hidden state for its predicted number of frames. Pitch and energy are predicted per phoneme, quantized into 256 bins, embedded and added to the hidden states.
Decoder	6 feed-forward Transformer blocks and a linear projection to 80 mel bins
PostNet	5 convolutional layers that refine the mel-spectrogram
Speaker embedding	The reference multi-speaker embedding, added to the encoder output. Used for English only.
Loss	Mel MAE + PostNet mel MAE + pitch MSE + energy MSE + log-duration MSE
Optimizer	Adam (β = 0.9, 0.98) with the reference warm-up and step-decay schedule
Vocoder	Pretrained HiFi-GAN `universal` generator, not retrained

During training the variance adaptor receives the ground-truth duration, pitch and energy. At inference it uses its own predictions.

1.2 Changes to the reference code (both languages)

File	Problem found	Change
`model/modules.py`	Energy prediction was scaled by the pitch control value	Uses the energy control
`audio/stft.py`, `audio/audio_processing.py`	Hardcoded `.cuda()` calls; `librosa` calls removed in current versions	Device-independent; keyword arguments for `librosa`
`preprocessor/preprocessor.py`	MFA 3 writes silence as unlabelled intervals, which were dropped. When a TextGrid was missing, statistics reused the previous clip's values. Durations and frame counts could disagree by rounding.	Silences kept and mapped to `sp`; statistics updated only for processed clips; rounding differences trimmed (up to 2 frames, otherwise the clip is rejected); sample rate passed explicitly
`utils/model.py`	Checkpoints only loaded on the training device; vocoder paths depended on the working directory	`map_location`; paths taken from the package location
`model/optimizer.py`	The learning rate was saved as a NumPy scalar, which PyTorch 2.x refuses to load safely	Stored as a plain float
`text/symbols.py`, `text/telugu.py`	No Telugu symbols	Separate Telugu symbol table, selected with `FASTSPEECH2_SYMBOLS=telugu`; the English table, and therefore the English embedding size, is unchanged

1.3 New code

Each language folder has its own `preprocess.py`, `train.py`, `inference.py` and `config.yaml`, as the assignment requires.

Shared training and inference behaviour (in the English scripts; the Telugu scripts reuse them):

- **Batching:** batches are limited by total padded frames, at most 6,000, so a 6 GB GPU never runs out of memory on long clips.
- **Robustness:** out-of-memory batches are skipped and logged, non-finite losses are rejected, and checkpoints are written atomically so a run can be resumed exactly.
- **Outputs:** loss logs go to CSV. Inference writes the audio, plus duration, pitch and energy plots and data, for each sentence.
- **Pitch-range control:** inference can widen the predicted pitch contour around each sentence's own mean. A second pass feeds the widened contour back as pitch targets, with the first pass's durations, so timing is unchanged. This uses FastSpeech 2's built-in ability to take pitch targets.

English-specific (`English/`)

- `text_frontend.py`, used for alignment and inference:
  - normalization of times, dates, years, ordinals, decimals, percentages, currency and long numbers;
  - lexicon lookup keeping all pronunciation variants, with `g2p_en` for unknown words;
  - part-of-speech choice for heteronyms;
  - spelled-out variants for acronyms.
- **Pronunciation choice from the training alignments.** When a word has several dictionary pronunciations, inference uses the one the aligner found most often in the training recordings (`preprocess.py pronunciations` records them). The model is then asked for the forms it actually heard. For example, "twenty" and "parliamentary" keep their T, and "during" is "dyuring", as these speakers say it.
- `select_clean_split.py` chooses the data (Section 2.1).
- `make_group_variant.py` links each recording's features under its demographic group for conditioning.
- `rank_voices.py` chooses the inference voice by speech-recognition error on validation sentences (Section 4.2).
- **Evaluation:** `evaluate_asr.py` and `evaluate_mcd.py` work for both languages (Section 4.2).

Telugu-specific (`Telugu/`)

- `telugu_frontend.py`, a rule-based phonemizer for Telugu script:
  - inherent vowel, vowel signs, virama, homorganic anusvara, visarga;
  - 46 phonemes, with retroflexes marked `R`;
  - Telugu number reading, including thousands, ordinals with ఒ, and clock times.
- `inspect_dataset.py` measures every recording from its audio header.
- `preprocess.py` selects the split and trains a Montreal Forced Aligner acoustic model, because no pretrained Telugu model exists.

Tests. `English/tests/` and `Telugu/tests/` hold 20 unit tests, covering:

- the energy-control fix;
- the text frontends, including the pronunciation choice;
- TextGrid parsing;
- vocoder settings;
- frame-limited batching;
- the separation of the two symbol tables.

2. Datasets, preprocessing, split and training configuration

2.1 Data selection and subsampling

English: AI4Bharat Svarah. 6,656 recordings (about 9.6 h) of Indian-accented English from 117 documented speakers, as 16 kHz mono WAV, with gender, age group, primary language, state and district. The Hugging Face release has no verified speaker IDs.

1. **Candidates.** Every recording of 1 to 20 s, with repeated transcripts removed: 5,160 recordings, 8.464 h. Excluded: 1,325 clips outside the length limits and 171 repeated transcripts.
2. **Recording quality.** After forced alignment (Section 2.3), each candidate gets a speech-to-silence energy ratio: the mean energy of its phone intervals over the mean energy of its silence intervals, in dB, on the source audio. 22 clips had no silence interval to measure against, leaving 5,138.
3. **Selection.** Clips were taken from the highest ratio downwards until reaching 6 h. The lowest kept ratio is 13.88 dB.
4. **Split.** The 6 h were shuffled with seed 42; the first 1 h became validation and the rest training.

Why the cleaner recordings were selected. The first English model used a plain random 5 h + 1 h split and was hard to understand.

- **Measured noise:** that random selection had a median ratio of 21.0 dB, with 46% of clips under 20 dB. The Telugu data had a median of 33.5 dB, with 2% under 20 dB.
- **What the ratio is:** an estimate, not a calibrated SNR. A quiet speaker also lowers it.
- **Effect of the cleaner selection:** it raised the training median to 23.2 dB and cut the share under 20 dB to 32%.

The random split is kept in `English/data/splits/` for reference.

Telugu: IndicTTS Telugu (`SPRINGLab/IndicTTS_Telugu`). 8,576 recordings (36.7 h):

- female class: 23.9 h at 16 kHz;
- male class: 12.8 h at 48 kHz.

The selection, with seed 42:

1. **Female class only**, so all training hours go to one voice. In a sample of 200 clips, the median F0 per clip formed one narrow distribution (176.5 to 240.4 Hz from the 1st to the 99th percentile). This is consistent with a single speaker, but no speaker ID confirms it.
2. **1 to 20 s per clip**, which removed 1,821 long, paragraph-style recordings.
3. **Repeated transcripts removed** (15).
4. **Random sampling:** 1 h for validation first, then 5 h for training.

2.2 Train/validation split

Language	Split	Recordings	Selected hours	Usable recordings	Usable hours after silence trimming
English	Training	3,162	4.9995	3,162	4.684
English	Validation	659	1.0012	659	0.937
Telugu	Training	1,576	5.0044	1,537	4.369
Telugu	Validation	297	1.0018	289	0.882

- **No overlap.** No recording is in both sets of either language. This was checked on the manifests and again on the final metadata.
- **Not speaker-exclusive.** The same English speakers appear in both sets: all 105 validation demographic groups also occur in training. The Telugu data is probably one voice. Validation loss therefore does not measure performance on new speakers.
- **English gender balance:** training has 1,663 female and 1,499 male clips; validation has 335 female and 324 male.
- **Why usable Telugu audio is lower:** the Telugu speaker pauses a lot. Silence takes a median of 32.6% of each clip, and leading and trailing silence is trimmed. 47 Telugu clips also failed alignment (Section 2.3).

2.3 Preprocessing

Audio. The selected audio is 16 kHz, and the HiFi-GAN vocoder expects:

- 22,050 Hz;
- an FFT size and window of 1024 and a hop of 256;
- 80 mel bins, from 0 to 8,000 Hz.

The audio was therefore resampled to 22.05 kHz (`soxr_hq`), not relabelled. A 16 kHz recording has no content above 8 kHz, which is exactly the top of the mel range. The original 16 kHz files are kept byte-for-byte for alignment. Every selected recording exported without errors.

Text to phonemes

Step	English	Telugu
Normalization	Numbers, dates, times, ordinals, decimals, percentages and currency become English words; 12 November becomes "twelfth November"	Numbers become Telugu words (2026 → రెండు వేల ఇరవై ఆరు, 12వ → పన్నెండవ, 10:45 → పది నలభై ఐదు); no training transcript contains digits
Phonemes	ARPAbet with stress, from the LibriSpeech lexicon. 1,498 of 7,849 word types were unknown and got `g2p_en` predictions, many of them Indian names.	Rules on the script. A consonant carries an inherent /a/ unless a vowel sign or virama follows; virama gives clusters and doubled consonants (పట్టు → `p a tR tR u`); anusvara becomes the nasal matching the next consonant.
Symbol table	Reference ARPAbet table (360 symbols)	New table of 50 symbols

Forced alignment. Both languages used Montreal Forced Aligner 3.4.3.

	English	Telugu
Acoustic model	Pretrained `english_us_arpa` (American English), beam 100, retry beam 400	None existed, so one was trained with `mfa train` on the selected data (11,840 s on CPU)
Speaker adaptation	By demographic group (gender, age group, language, state, district: 110 groups), as no speaker IDs exist	Not needed: one voice
Result	All 5,160 candidates aligned	1,826 of 1,873 aligned; the other 47 (2.5%) were dropped from both sets
Sanity check	The English dictionary gives every pronunciation variant, and the aligner's choices are recorded per word (6,093 word types in training)	Median aligned vowel lengths follow Telugu vowel length: a/aa 50/100 ms, i/ii 60/110 ms, u/uu 50/100 ms, e/ee 60/110 ms

Features. The reference `Preprocessor.process_utterance` extracts, per clip:

- an 80-bin mel-spectrogram;
- pitch, using WORLD DIO with StoneMask, interpolated over unvoiced frames;
- energy, the L2 norm of each STFT frame;

- the frame duration of each phoneme, from the TextGrid.

Pitch and energy are averaged per phoneme and z-score normalized. **The normalization statistics come from training data only;** the reference code computes them over all data.

Training-data statistic	English	Telugu
Pitch mean / standard deviation (Hz)	178.6 / 58.0	212.6 / 41.2
Energy mean / standard deviation	28.7 / 27.0	36.3 / 23.9

Checks on every clip:

- mel shape;
- equal numbers of phonemes and durations;
- durations summing to the mel length;
- finite values;
- no phoneme dropped by the symbol table.

No clip failed these checks in either language.

2.4 Training configuration

The configurations are `English/config.yaml` and `Telugu/config.yaml`.

Setting	Reference (LJSpeech)	English	Telugu
Speaker embedding	Off	On: 109 demographic groups	Off
`max_seq_len` (frames)	1000	2700	2000
Symbols	ARPAbet	ARPAbet (360)	Telugu (50)
Text cleaners	`english_cleaners`	`english_cleaners`	none
Batch	16 utterances	16 utterances, at most 6,000 padded frames	same as English
Warm-up steps	4,000	4,000	4,000
Learning-rate decay (×0.3) at	300k, 400k, 500k	10k, 14k, 18k	25k, 35k, 45k (not reached)
Steps	900,000	20,000	25,000 of 50,000 planned
Gradient clipping	1.0	1.0	1.0
Checkpoint interval	100,000	5,000	2,500
Parameters		36,057,665	35,592,001
Inference voice / pitch range		`Male-30-45-Maithili-Bihar-Darbhanga` / ×1.3	single voice / ×1.0

- **Why `max_seq_len` was raised:** 1,000 frames is 11.6 s, which would truncate longer clips.
- **Why English uses demographic groups:** without voice conditioning, the model averages many speakers' pitch and voice quality (Section 3.2).

Hardware: an NVIDIA GeForce RTX 4050 Laptop GPU (6 GB), on Windows 11 with Python 3.10 and PyTorch 2.11.0 (CUDA 12.8). Peak allocated GPU memory was 1.59 GB for both models. No English batch ran out of memory.

The Telugu crash. At step 10,398 a batch ran out of GPU memory, and the recovery step itself failed, which ended the run. Training resumed from the step-10,000 checkpoint, losing 398 steps. The cause was not established. Recovery now tolerates the failure, and checkpoints are saved more often.

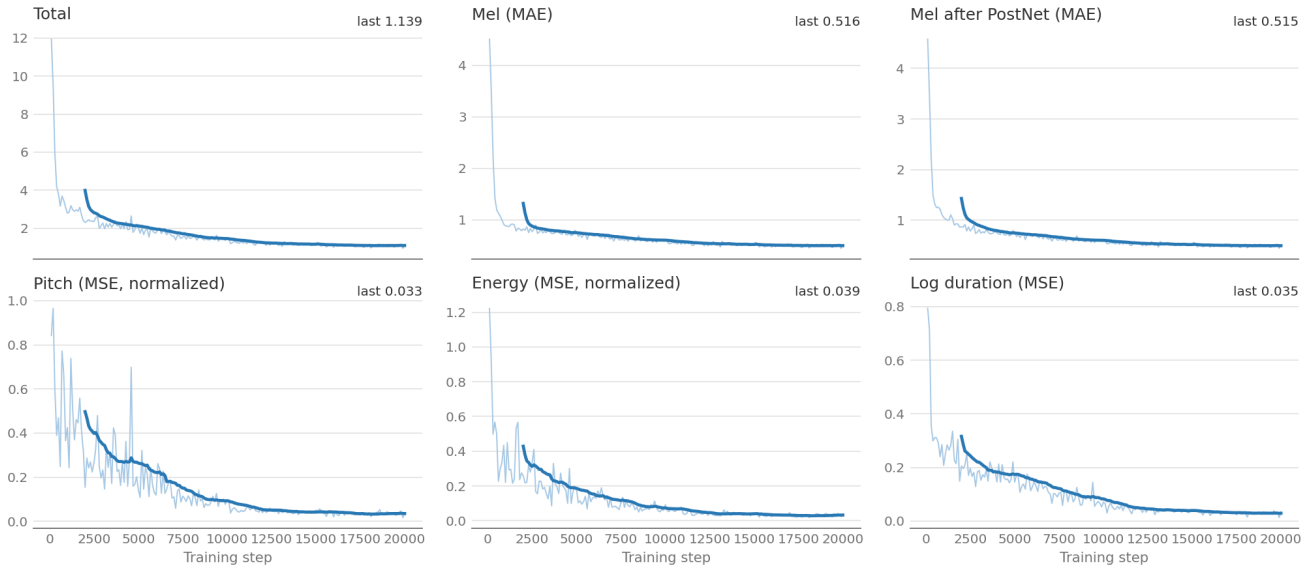
3. Training and validation loss analysis

Validation losses are computed with the ground-truth durations, pitch and energy fed to the decoder, so they don't measure free-running synthesis. Section 4.2 measures that.

3.1 English

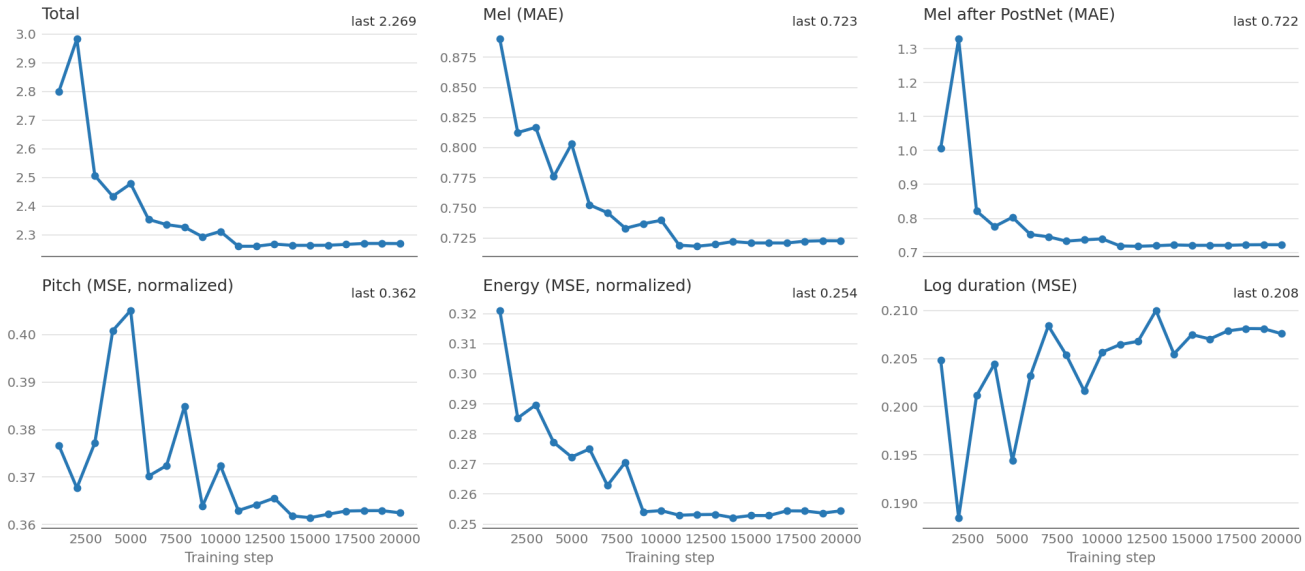
English FastSpeech 2: training loss

200 logged batches to step 20000. Pale: logged values. Dark: moving average of 20 logged values.



English FastSpeech 2: validation loss

20 evaluations to step 20000, averaged over the validation set using ground-truth durations.



Step	Validation total	Mel	PostNet mel	Pitch	Energy	Log duration
2,000	2.983	0.812	1.329	0.368	0.285	0.188
5,000	2.478	0.803	0.803	0.405	0.272	0.194
10,000	2.312	0.740	0.740	0.372	0.254	0.206
12,000 (lowest total)	2.260	0.718	0.718	0.364	0.253	0.207

Step	Validation total	Mel	PostNet mel	Pitch	Energy	Log duration
20,000 (final)	2.269	0.723	0.722	0.362	0.254	0.208

The last logged training batch (step 20,000) had total 1.139, mel 0.516, PostNet mel 0.515, pitch 0.033, energy 0.039 and log duration 0.035.

- **Convergence.** Training loss falls smoothly, and its moving average levels off after about step 12,000, following the learning-rate decay at step 10,000. Validation total drops to 2.26 just after that decay and then stays within 0.01.
- **Instability.** Training pitch loss is spiky in the first 5,000 steps, with single batches up to about 0.7 against an average near 0.3. Validation pitch and mel move irregularly until step 10,000. The PostNet loss at step 2,000 is high (1.329) while the PostNet is still learning.
- **Overfitting.** Validation stops improving after step 12,000, while training loss keeps falling to half the validation value. Validation log duration is lowest at step 2,000 (0.188) and creeps up to 0.208, so the duration predictor starts fitting the training clips early.

3.2 How the English model was arrived at

Two earlier English models were trained and evaluated; they are not part of the submission.

Run	Data	Conditioning	Steps	Validation total (lowest / final)	Pitch	Listening
1	Random 5 h + 1 h	Gender (2 classes)	50,000	3.477 at step 13,000 / 3.510	0.710	Hard to understand
2	Same as run 1	Demographic groups (110)	20,000	2.447 at step 11,000 / 2.476	0.422	Better
3 (final)	**Cleanest 6 h**	**Demographic groups (109)**	**20,000**	**2.260 at step 12,000 / 2.269**	**0.362**	**Clearest of the three**

Runs 1 and 2 share a validation set. Going from gender to group conditioning cut every validation loss. Run 1 also overfitted: validation stopped improving by step 13,000, but training continued to 50,000.

Run 3 is scored on a different, cleaner validation set, so its lower losses partly reflect an easier test. Each run was therefore also compared with predictors that ignore the text, on its own validation set (normalized MSE):

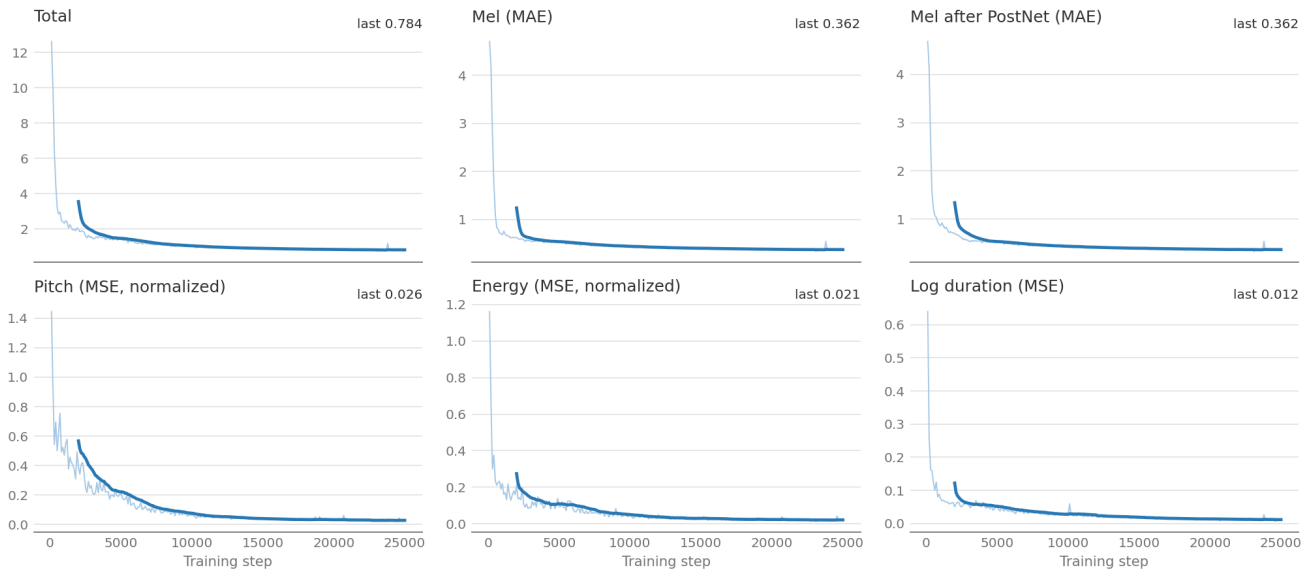
Term	Run 1	Gender mean	Run 2	Group mean	Run 3	Group mean
Pitch	0.710	0.656	0.422	0.467	0.362	0.385
Energy	0.802	1.065	0.311	0.788	0.254	0.673
Log duration	0.253	0.358	0.222	0.354	0.208	0.317

- **Run 1's pitch predictor was worse than simply using each gender's average pitch.** Most pitch variation in Svarah is between voices, which two gender classes cannot capture.
- **Group conditioning gives the predictor that information,** and runs 2 and 3 beat their group-mean baselines.
- **Cleaner data made the speech easier to understand,** but it did not improve the predictions relative to these baselines.

3.3 Telugu

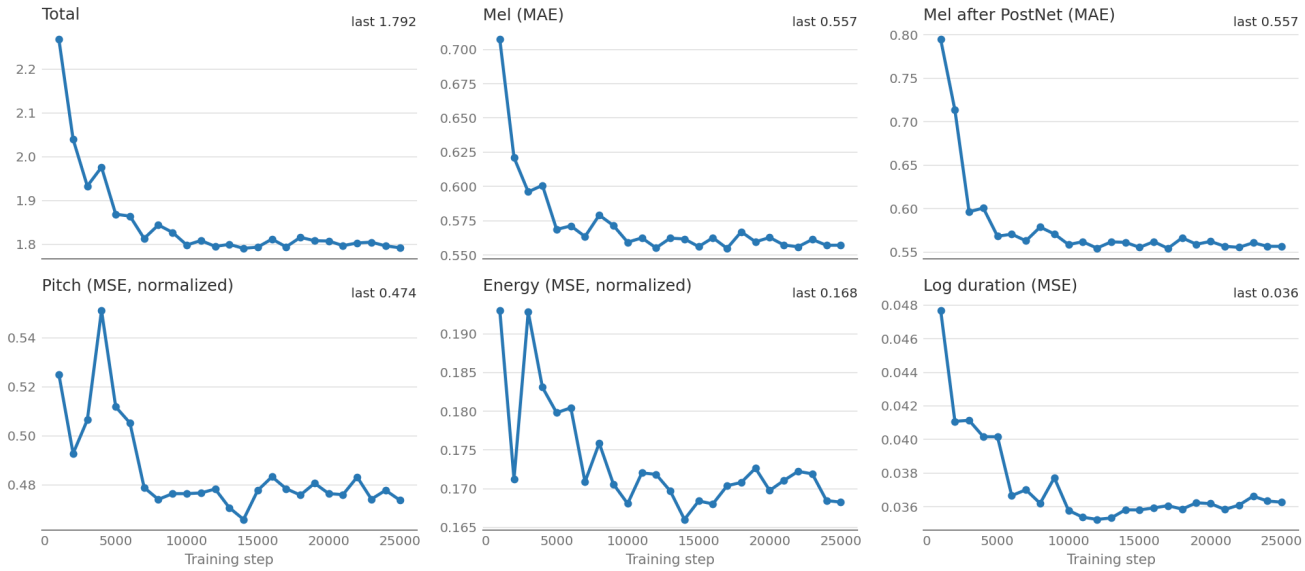
Telugu FastSpeech 2: training loss

250 logged batches to step 25000. Pale: logged values. Dark: moving average of 20 logged values.



Telugu FastSpeech 2: validation loss

25 evaluations to step 25000, averaged over the validation set using ground-truth durations.



Step	Validation total	Mel	PostNet mel	Pitch	Energy	Log duration
2,000	2.040	0.621	0.713	0.493	0.171	0.041
10,000	1.798	0.559	0.559	0.476	0.168	0.036
14,000 (lowest total)	1.791	0.562	0.561	0.466	0.166	0.036
20,000	1.808	0.563	0.562	0.476	0.170	0.036
25,000 (final)	1.792	0.557	0.557	0.474	0.168	0.036

The last logged training batch (step 25,000) had total 0.784, mel 0.362, PostNet mel 0.362, pitch 0.026, energy 0.021 and log duration 0.012.

- **Convergence.** Training loss falls smoothly for the whole run. Validation loss improves quickly up to step 10,000 and then stays within 0.02. Training was therefore stopped at step 25,000 of the planned 50,000, before the first learning-rate decay.
- **Instability.** One training batch near step 23,800 has a mel-loss spike, and the next logged batches are normal. The out-of-memory crash at step 10,398 interrupted the run but did not change the curves, because training resumed from step 10,000 and the logs were trimmed to that step.
- **Overfitting.** The final training total is less than half of validation, so the model fits training clips more closely than new ones. Unlike English, validation pitch and duration do not rise.

4. Generated speech: pronunciation and quality

All test files are 22.05 kHz mono WAV with no clipped samples, in `Results/English/test_generated_audios/` and `Results/Telugu/test_generated_audios/`. The sentence text is used exactly as given. The PDF prints E3 and T3 with a gap inside a word ("rhyth m", అడ విలో); both are read as the whole word.

4.1 Observations for each test sentence

The listening notes are the author's.

- **Telugu:** the notes are for the submitted files.
- **English:** the notes were made on the first version of the outputs, which used the largest female group's voice, the first dictionary pronunciation of each word, and an unchanged pitch range. Those notes led to three inference changes: pronunciation from the training alignments, a voice chosen by speech recognition, and pitch range ×1.3. The author then compared candidates side by side and chose the submitted version as the clearest.

The "Whisper heard" column is the transcript of the submitted file (Section 4.2).

ID	Pronunciation given to the model	Listening notes	Whisper heard
E1	"antidisestablishmentarianism" is not in the lexicon. Its entry is built from the lexicon's "anti" and "disestablishment" plus the "-tarianism" ending: `AE2 N T IY0 D IH0 S IH0 S T AE2 B L IH0 SH M AH0 N T EH1 R IY0 AH0 N IH2 Z AH0 M` (g2p gave "an-tie-dis-uh-..."). "parliamentary" now keeps its T, as aligned in training.	Hard to follow; the long word could not be caught clearly; monotonous; "parliamentary" mispronounced. Fixed in the final version: the T in "parliamentary", the long word's entry, and a wider pitch range.	"the anti-dissist-arist mentalism movement in considerable momentum during the parliamentary proceedings."
E2(a)	"conduct" as a noun, stressed on the first syllable: `K AA1 N D AH0 K T`	Clear and understandable, but monotonous. Pronounced "CON-duct"; the stress is correct for the noun, but the word sounded slightly off.	"The orchestra's conduct during the ceremony was praised by the audience."
E2(b)	"conduct" as a verb, stressed on the second syllable: `K AH0 N D AH1 K T`. The reference frontend would have used the noun form.	Mostly clear but robotic. Pauses and slight glitches.	"The researchers will conduct an experiment to evaluate the proposed method."
E3	"rhythm" `R IH1 DH AH0 M`	Clear, but "unexpectedly" was not pronounced clearly	"The rhythm of the orchestra changed unexpectedly."
E4	Spoken as "ten forty five A M on twelfth November twenty twenty six". "AM" is read as the letters. The romanized Telugu "andaroo" (అందరూ, "everyone") is set by hand to `AH1 N D AH0 R UW1`, the nearest English phonemes.	Clear. "andaroo" was pronounced properly. Some robotic background noise in between. "10:45" was not fully clear.	"The meeting will start at time 45 a.m. on 12th November, 2026. So, Anjaro should arrive on time."
T1	Loanword న్యూరోఫార్మకోలాజికల్ read from its spelling: `n y uu r oo ph aa r m a k oo l aa j i k a l`. ఫ is aspirated `ph`.	Very clear. "neuro" pronounced like Telugu speech. No changes needed.	Devanagari: निवूरो भार्म कोलाजि कल्परि सोदनलों तनवूशदाल प्रभावान नी विसलेषिचारू
T2(a)	పట్టు (grip) = `p a tR tR u`	Good. The pause between తాడుపై and అతని was natural.	Devanagari: ताड पयतनी पत्तू चाला बलंगा हूंदी

ID	Pronunciation given to the model	Listening notes	Whisper heard
T2(b)	పట్టు (silk), the same input `p a tR tR u`	Every word pronounced properly, including the stress on పట్టు and the lengthening in చీర	Devanagari: आमे पट्ट्ट्ट्ट्ट्ट्ट्ट्ट्ट्ट्ट्ट्टि
T3	అడవిలో; నిశ్శబ్దంగా has a doubled `sh sh`	నిశ్శబ్దంగా, a difficult word, was pronounced properly; the rest was perfect	Devanagari: पूलि अदविलो निस सब्दंगा नदिचिन्दि
T4	Numbers as Telugu words: రెండు వేల ఇరవై ఆరు, పన్నెండవ, పది నలభై ఐదు	Every date and time pronounced properly in Telugu	Devanagari: इसमा वेसम 2026 नवम बर्पन नदवत एदी उदयम पदिनलब है एदु गंतलक। प्रारम्भम हो तुंदी, समयानी की सिद्धंगा उंचंदी

4.2 Objective measures

Speech recognition (intelligibility). OpenAI Whisper (`small`, beam 5, temperature 0) transcribed:

- the test sentences;
- 100 randomly chosen validation clips per language (seed 42), each in two versions: the real recording, and the model's speech for the same text, using its own durations, pitch and energy.

Transcribing the real recordings shows how well Whisper itself copes with each language and accent, so the synthetic scores can be read against that reference. English text is normalized with Whisper's English normalizer; Telugu only loses punctuation. Word error rate (WER) and character error rate (CER, spaces ignored) are summed over the clips.

English	WER	CER
Validation, real recordings	7.2%	2.7%
Validation, synthesized	20.3%	9.9%
Test sentences (58 words)	17.2%	6.3%

On English, the synthetic speech loses intelligibility against the real recordings, but most words are still recognized. The test set is small, so single words move its score a lot. Whisper on the GPU is also not perfectly repeatable: transcribing the same E3 file twice gave "rhythm" once and "item" once. Differences of a few points are therefore within noise.

Telugu: no usable score.

- **Whisper fails on the real recordings too.** On the 100 validation clips, `small` produced over 100% word errors even for the real recordings: mostly long repetitions of one syllable, on clips of 10 to 20 s. The cause is Whisper's weak Telugu, not the model, so no Telugu ASR score is reported.
- **The short test sentences are informative.** Whisper wrote them phonetically in Devanagari (Hindi) script. T3, for example, came back as पूलि अदविलो निस सब्दंगा नदिचिन्दि ("puli adavilo nis sabdanga nadichindi"), a close sound-by-sound match for పులి అడవిలో నిశ్శబ్దంగా నడిచింది. These transcripts are shown in Section 4.1 as qualitative evidence only.
- **A fairer check** would need a recognizer trained for Telugu.

What the English inference changes did.

English test sentences	WER
First version: largest female group, first dictionary pronunciation, pitch ×1.0	48%

English test sentences	WER
Pronunciation from training alignments, voice chosen by Whisper	22%
Same, plus pitch range ×1.3 (submitted)	17.2%

How the voice was chosen. For the 10 demographic groups with the most training clips, 30 validation sentences were synthesized in each voice and transcribed (`English/rank_voices.py`). The voice with the lowest WER was chosen. The test sentences played no part in this.

Voice (group)	Training clips	WER
Male-30-45-Maithili-Bihar-Darbhanga	74	**9.9%**
Male-18-30-Odia-Odisha-Sundergarh	86	10.4%
Female-30-45-Nepali-West-Bengal-Kalimpong	96	10.9%
Female-18-30-Malayalam-Kerala-Pathanamthitta	92	11.5%
Female-18-30-Nepali-West-Bengal-Darjeeling	60	12.5%
Female-45-60-Urdu-Maharashtra-Satara	65	12.8%
Male-30-45-Urdu-Gujarat-Ahmedabad	106	13.1%
Female-30-45-Kashmiri-Jammu-Kashmir-Srinagar (first default)	107	13.9%
Female-18-30-Bodo-Assam-Kokrajhar	71	17.1%
Male-18-30-Nepali-West-Bengal-Kalimpong	76	19.5%

The top three voices are within about 1 percentage point, which is inside the noise of a 30-sentence test.

Why a wider pitch range. Models trained on mean-squared error predict an averaged, over-smoothed pitch contour, and the author heard the English output as monotonous. Widening the contour ×1.3 around each sentence's mean raised the pitch spread of the test sentences from 16 to 21 Hz (standard deviation across phonemes), and did not hurt recognition. Timing is unchanged.

Mel-cepstral distortion (MCD) compares generated speech with the real recording of every validation clip:

- **Analysis:** WORLD analysis, 25 coded mel-cepstral coefficients without c0, and DTW alignment, because the model's timing differs from the recording's.
- **Two conditions:**
  - the full system;
  - the vocoder alone, re-synthesizing each clip's real mel-spectrogram. This is the floor the vocoder allows.
- **Scale:** the values are comparable within this report only. Other papers use different analysis settings.

	Vocoder only (floor)	Full system	Gap added by the acoustic model	Synthesized / real length (median)
English (659 clips)	10.14 dB	11.66 dB	1.52 dB	0.94
Telugu (289 clips)	6.86 dB	8.29 dB	1.42 dB	0.98

The acoustic model adds a similar distortion in both languages. English starts from a far higher vocoder floor. The pretrained vocoder reproduces the noisy, varied Svarah recordings much less faithfully than the clean Telugu studio voice. This matches the "robotic background noise" heard in E4, and points to fine-tuning the vocoder as the next improvement (Section 6.5).

5. Variance adaptor analysis: duration, pitch and energy

Plots for every test sentence are in `Results/<language>/variance_adaptor_plots/{duration,pitch,energy}/`, and the plotted values are in `<language>/logs/inference/`.

- **Inference-time values only.** Every value is an inference-time prediction. For English, the pitch is shown after the ×1.3 widening, because that is what the decoder received.
- **Pitch on voiceless sounds.** Pitch and energy are per phoneme. Training pitch was interpolated across unvoiced frames, so pitch on voiceless consonants is not physical pitch.
- **Duration steps.** Durations come in steps of 11.6 ms, one mel frame.

5.1 Patterns shared by all sentences

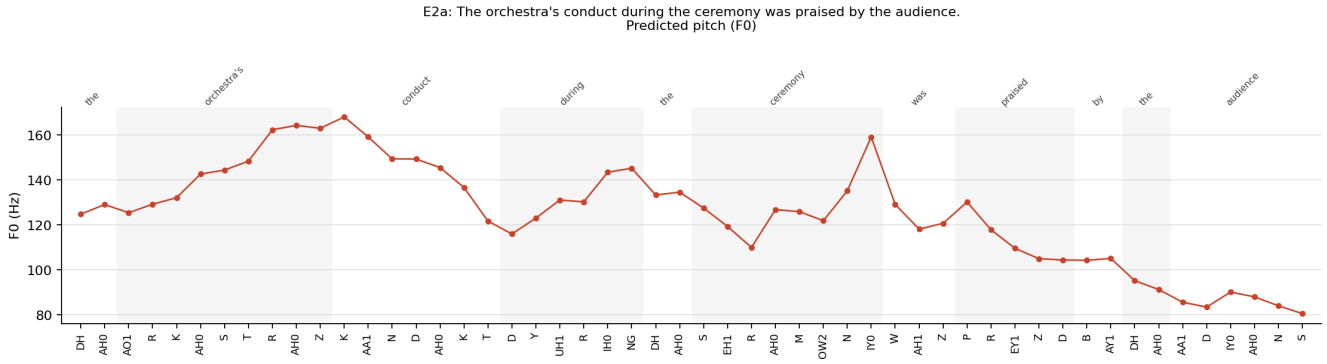
ID	Mean phoneme duration (ms)	F0 range (Hz)	Mean F0 (Hz)	F0 of first → last phoneme (Hz)	Mean energy, vowels / consonants
E1	65.7	80–166	123	121 → 80	19.5 / 10.5
E2(a)	76.4	80–168	125	125 → 80	17.4 / 9.2
E2(b)	70.9	91–176	127	94 → 101	16.7 / 7.9
E3	71.9	96–178	119	104 → 104	18.2 / 7.7
E4	78.5	76–176	121	112 → 77	16.1 / 8.3
T1	77.4	163–312	210	239 → 163	48.4 / 31.9
T2(a)	80.5	168–336	228	286 → 169	52.5 / 27.4
T2(b)	72.8	163–335	227	225 → 168	55.3 / 28.7
T3	70.8	164–338	223	273 → 170	56.6 / 31.9
T4	71.7	158–296	211	268 → 165	46.8 / 30.6

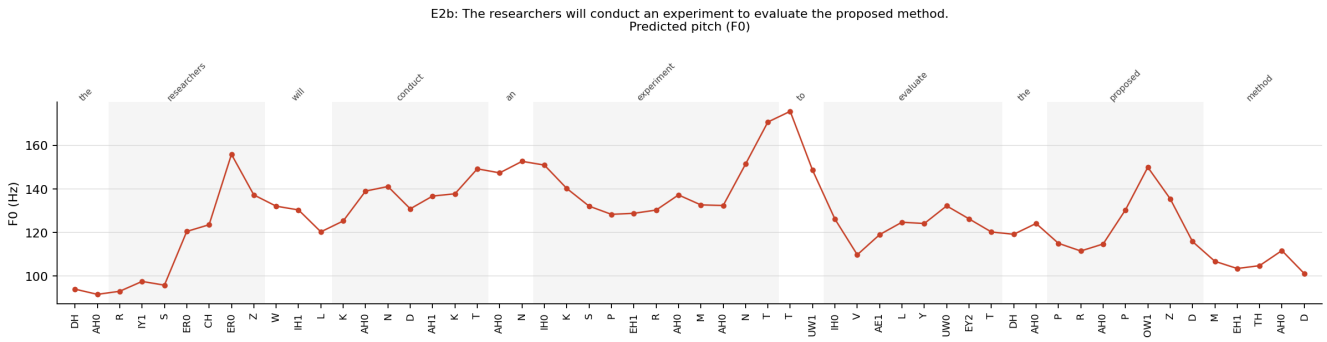
- **Pitch.** It falls across most sentences (declination), as in natural declarative speech. In the English male voice, E1, E2(a) and E4 end near 80 Hz. E2(b) and E3 end flat or slightly rising, and their final words ("method", "unexpectedly") stay low throughout, which matches "unexpectedly" being heard as unclear. Telugu, a female voice, has a higher and wider range.
- **Energy.** On vowels it is roughly twice that on consonants. Voiceless stops and fricatives are the quietest sounds.
- **Duration.** Sounds lengthen at the ends of phrases: `Z` in "proceedings" (128 ms), `S` in "audience" (139 ms), the final `IY0` of "unexpectedly" (186 ms), and the final `i` of ప్రారంభమవుతుంది (151 ms).
- **No phoneme** was predicted to have zero duration.

5.2 English sentences

E1.

- **The long word** takes 1,846 ms. Its highest pitch (166 Hz, the sentence maximum) falls on the unstressed `IH0` of "-es-". The primary stress `EH1` ("-tar-") gets 125 Hz but one of the word's longer vowels (93 ms). So duration partly marks the main stress, and pitch does not.
- **"parliamentary"** takes 697 ms and now includes its `T` (46 ms).
- **"proceedings"** ends the sentence at the lowest pitch (80 to 101 Hz), with long `NG` (116 ms) and `Z` (128 ms).

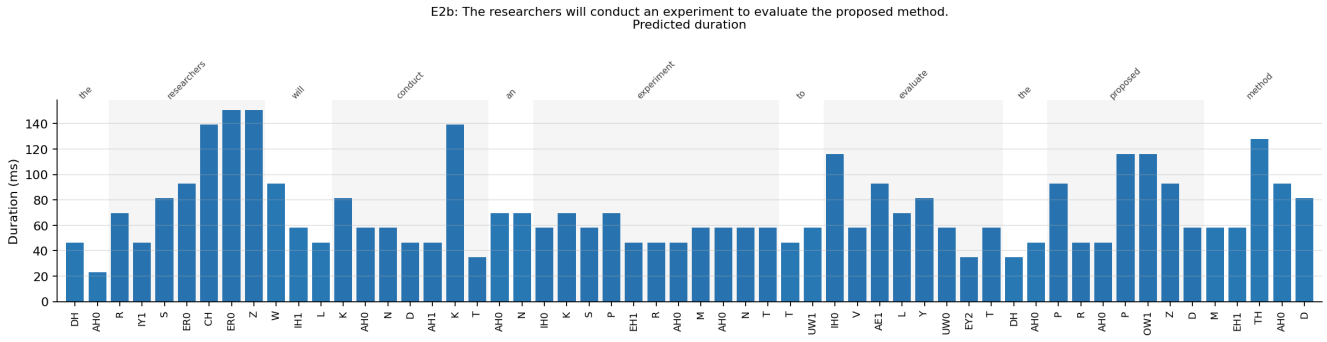




E2: the heteronym "conduct".

Sentence	Stressed vowel	Unstressed vowel	Stressed is longer?	Louder?	Higher?
E2(a), noun	`AA1` : 70 ms, 159 Hz, energy 19	`AH0` : 70 ms, 146 Hz, energy 19	Equal	Equal	Yes
E2(b), verb	`AH1` : 46 ms, 137 Hz, energy 19	`AH0` : 58 ms, 139 Hz, energy 19	No	Equal	No

- **The noun** carries the sentence's pitch peak: 168 Hz at its first `K`, then 159 Hz on the stressed `AA1`. That makes "conduct" the accented word of E2(a), with stress on its first syllable, as the noun requires.
- **The verb** shows no stress cue in duration, pitch or energy. Its stress comes only from the vowel qualities the frontend supplies (`AH0` then `AH1`), which Whisper still recognized as "conduct".
- **The final `K` of the verb** is long (139 ms), just before "an".



E3.

- **"rhythm"** takes 290 ms and rises to the sentence's highest pitch at its end (`AH0` 165 Hz, `M` 178 Hz), against 120 Hz on the stressed `IH1`.
- **"unexpectedly"** takes 917 ms, with very low energy (2 to 14) and low pitch (96 to 115 Hz) and a long final `IY0` (186 ms). That fits the author hearing it as unclear.

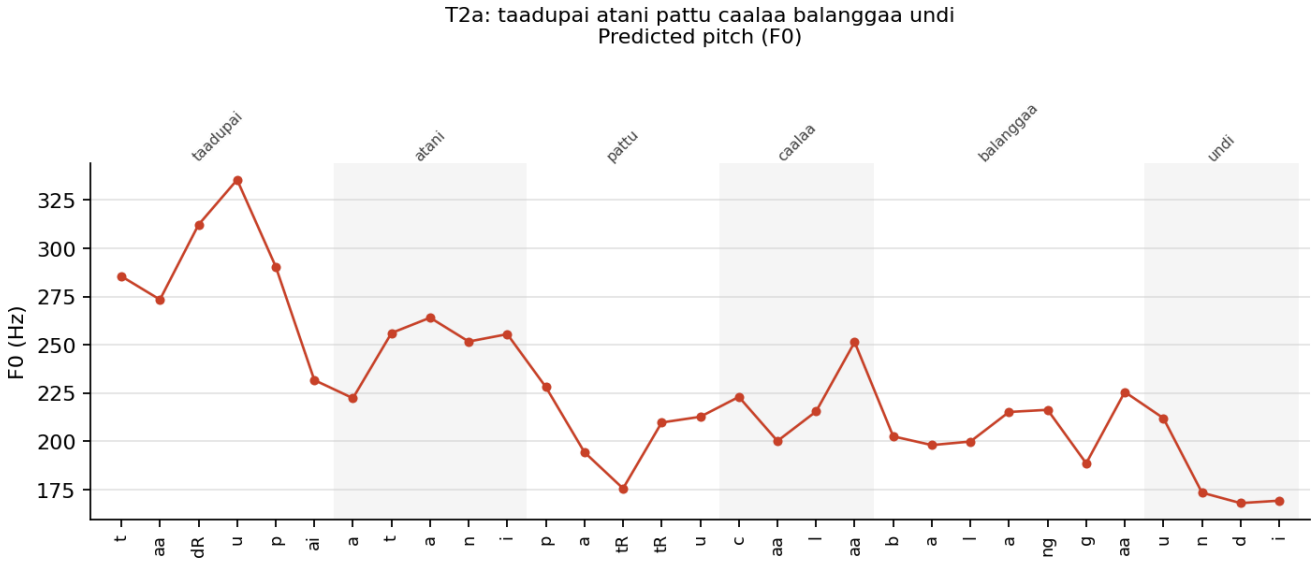
E4.

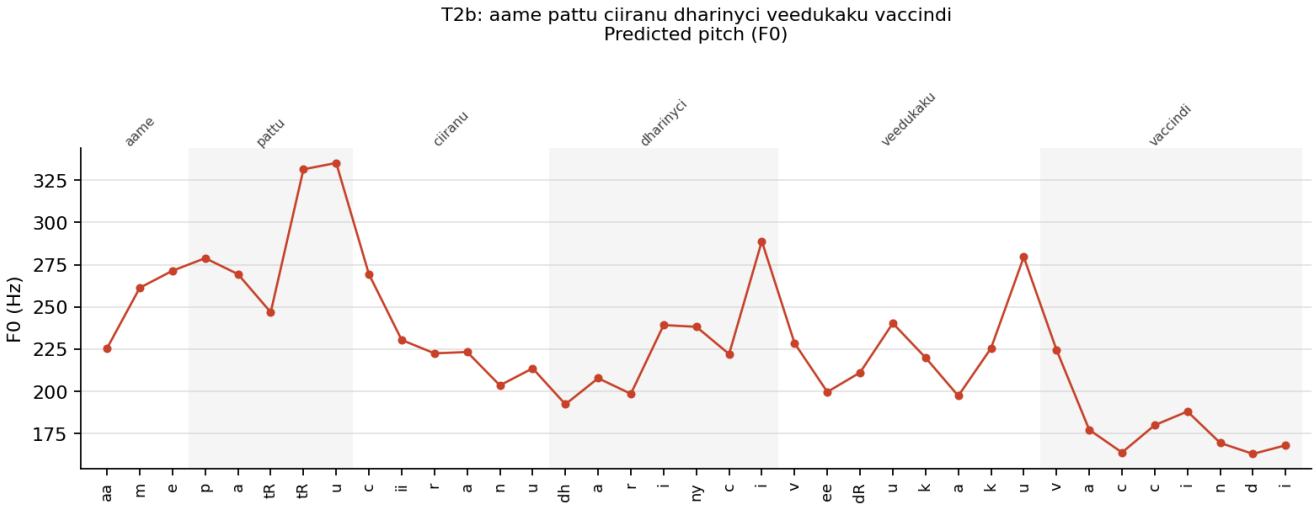
- **The pause.** The comma before "so" gives the only pause, 1,103 ms.
- **"five"** has the longest vowel (`AY1`, 197 ms).

- **"ten"** is very short (139 ms, with a 35 ms vowel). That fits "10:45" being heard as not fully clear.
- **"twenty twenty"** gets different timing for its two identical words (417 and 314 ms), so the predictions depend on context.
- **"andaroo"** takes 488 ms, and its final long `UW1` (93 ms) rises to 154 Hz. It was heard as pronounced properly.
- **The highest pitch** (176 Hz) falls on "meeting", at the start of the sentence.

5.3 Telugu sentences

T1. The loanword takes 1,602 ms and carries the sentence's highest pitch (312 Hz on `oo`) in its first syllables. The aspirated `ph` is long (128 ms) and quiet (energy 20), as an aspirated stop should be, not a fricative /f/. Pitch then falls to 163 Hz by the end of విశ్లేషించారు.





T2: the homograph పట్టు.

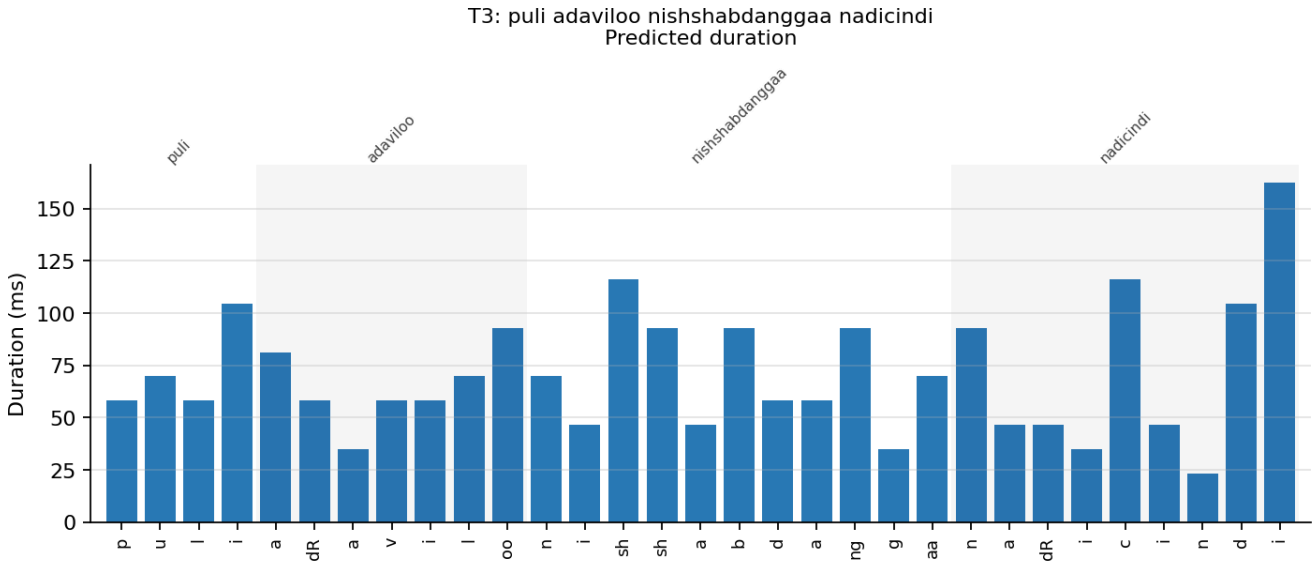
	T2(a), grip: తాడుపై అతని పట్టు చాలా బలంగా ఉంది	T2(b), silk: ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది
Position	Third word	Second word
Duration	337 ms	325 ms
F0 across `p a tR tR u` (Hz)	228, 194, 176, 210, 213	279, 269, 247, 331, 335

	T2(a), grip: తాడుపై అతని పట్టు చాలా బలంగా ఉంది	T2(b), silk: ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది
Energy on the final `u`	59	85

The input is identical, so the model cannot know which meaning is intended, and the duration barely changes. Pitch and energy do change with position:

- **T2(b):** the word directly follows a one-word subject and carries the sentence's pitch peak (335 Hz), like a phrase accent. The author heard this as stress on పట్టు.
- **T2(a):** it comes later, lower in the falling contour.

The difference is a reaction to sentence structure, not to meaning.



Vowel length and doubled consonants. Telugu vowel length is phonemic, and the predictions keep it. Median predicted durations across T1 to T4:

Pair	Short	Long
a / aa	46 ms (50 vowels)	104 ms (17)
i / ii	46 ms (24)	128 ms (3)
u / uu	46 ms (17)	110 ms (2)
e / ee	70 ms (3)	104 ms (5)

Long vowels are about twice as long as short ones, the same ratio as in the forced alignments. The `ii` of చీర in T2(b) is one of them, which matches the lengthening heard at చీర. Doubled consonants are longer than single ones, which usually take 35 to 93 ms:

- `tR tR` in పట్టు: 128 ms in T2(a) and 127 ms in T2(b);
- `sh sh` in నిశ్శబ్దంగా: 209 ms.

T3. అడవిలో takes 453 ms. Pitch rises on `i l oo` (279, 275, 266 Hz) after a low retroflex `dR` (194 Hz). The sentence's highest pitch (338 Hz) is on its first word, పులి.

T4. The number words take:

- రెండు వేల ఇరవై ఆరు: 290, 290, 372 and 209 ms;
- పన్నెండవ: 592 ms;
- పది నలభై ఐదు: 232, 464 and 255 ms.

Within them, long vowels are long: `ee` in వేలు is 104 ms against 46 ms for its `a`, and `aa` in ఆరు is 116 ms against 35 ms for its `u`. The comma gives the only pause, 488 ms. Before it, ప్రారంభమవుతుంది ends with a long final `i` (151 ms) at low pitch (189 Hz).

6. Comparison of the English and Telugu models

6.1 What differs in the implementation

	English	Telugu
Text to phonemes	Dictionary and g2p. About 19% of word types are predicted, so some pronunciations can be wrong. Variants follow what the speakers said in training.	Rules. Telugu spelling is close to pronunciation, so every word gets a consistent reading.
Aligner	Pretrained on American English, applied to Indian English	Trained on the same voice it aligns
Voices	Many speakers, grouped by demographics; inference voice chosen by speech recognition	One voice
Recording conditions	Varied: median speech-to-silence ratio 23.2 dB in the selected training data	Consistent studio audio: median 33.5 dB

6.2 Why Telugu is easier to understand

How much of the validation pitch error remains with each predictor (normalized MSE):

Predictor	English	Telugu
One global mean, ignoring the text	1.104	1.157
Mean of the conditioning label (group / single voice)	0.385	1.157
Each clip's own mean (an oracle, not achievable)	0.312	1.085
Model	**0.362**	**0.474**
Model's improvement over the label mean	6%	59%

- **Pitch.** In English, most variation is between voices and recordings: knowing the group removes most of the error, and even a clip's own average leaves little for the text to explain. In Telugu, almost all variation is inside each sentence, which is what a text-to-speech model learns to predict.
- **Timing.** Converted to everyday units, English phoneme durations are off by a typical factor of about 1.6, close to the 1.8 of a text-free guess. Telugu's are off by about 1.2. Wrong timing makes speech hard to follow even when each sound is right.
- **Vocoder and recordings.** The MCD results (Section 4.2) show the same thing from the signal side: the acoustic model's share of the distortion is similar in both languages, but English starts from a much worse vocoder floor.

The total losses (English 2.269, Telugu 1.792) come from different voices and validation sets, so they should not be compared directly.

6.3 Behaviour on the test-sentence challenges

Challenge	English	Telugu
Rare or very long word (E1, T1)	Built from dictionary parts. Duration partly marks the main stress; pitch does not. Still hard to catch by ear.	The rules give a reading for any spelling. Heard as natural, with a strong pitch accent. A loanword's /f/ becomes `ph`.
Same spelling, different role (E2, T2)	The frontend picks the stress from the part of speech. The noun is accented by pitch; the verb only by vowel quality. Both were recognized correctly.	Identical input. Pitch and energy follow the word's position; duration hardly changes. Meaning is not distinguished, though the stress was heard as natural.

Challenge	English	Telugu
Gap printed inside a word (E3, T3)	Read as "rhythm"	Spaces don't change Telugu phonemes, so the gap has no effect
Numbers, date, time (E4, T4)	Normalized to English words; "ten" is predicted very short and was not fully clear	Normalized to Telugu words and heard as pronounced properly
Word from the other language (E4 "andaroo", T1 loanword)	Nearest English phonemes; heard as pronounced properly	Read through Telugu phonemes; heard as Telugu-style speech
Phrasing	One pause at the comma (1,103 ms); pitch falls over most sentences	One pause at the comma (488 ms); pitch falls over each sentence

6.4 Limitations and failures

Limitations of both models

- **Validation measures:** neither split is speaker-exclusive, so validation doesn't measure generalization to new speakers. Validation losses also use ground-truth durations; the ASR and MCD measures cover free-running synthesis.
- **Training:** both models overfit after 10,000 to 14,000 steps, and both trained far shorter than the reference 900,000 steps. HiFi-GAN was not fine-tuned on either dataset.
- **Evaluation:** quality was judged by one listener, plus automatic measures. There was no MOS test.

English limitations

- **Intelligibility:** still below Telugu. It still sounds robotic in places, the long word in E1 is hard to catch, and the verb "conduct" has no prosodic stress cue.
- **Speaker groups:** demographic groups are not verified speakers and may mix several voices.
- **Data selection:** the cleaner selection favours loud, clean recordings, which changes the mix of voices and makes its validation set easier.
- **Alignment and pronunciation:** the American English aligner, and about 1,500 predicted word pronunciations, limit alignment and pronunciation accuracy.
- **Inference settings:** the voice and pitch range are settings chosen at inference. The voice was chosen on validation sentences. The pitch range was chosen by listening, and confirmed not to hurt recognition.

Telugu limitations

- **Voice:** a single voice, with no speaker control.
- **Unreviewed rules:** the phonemizer rules have not been reviewed by a linguist, and ఫ is always `ph`.
- **Training cut short:** stopped at step 25,000, before any learning-rate decay. 2.5% of the selected clips failed alignment.

Failures along the way

- English run 1 was hard to understand, and its pitch predictor did worse than a gender average.
- The first English inference version was monotonous, with dictionary forms that differed from what the speakers say.
- The Telugu run crashed once from an out-of-memory error and was resumed.

6.5 Possible improvements

- **English:**
  - Fine-tune HiFi-GAN on Svarah, since MCD shows the vocoder floor is the largest gap.

    - Recover real speaker labels, and train on fewer, cleaner voices or on denoised audio.
    - Adapt the aligner to Indian English, and extend the dictionary with Indian names.
    - Add a stress or accent input, so pitch follows lexical stress (E1, E2(b)).
    - Start from a model pretrained on a large English corpus.
- **Telugu:**
    - A loanword exception list (for example ఫ as /f/).
    - More of the 24 h female data.
    - Training through the planned learning-rate decay.
- **Both:**
    - Early stopping on validation loss, keeping the best checkpoint.
    - A speaker-exclusive evaluation.
    - A listening test with several listeners.

7. Reproducibility

7.1 Repository layout

Item	English	Telugu
Scripts	`English/{preprocess,train,inference}.py`	`Telugu/{preprocess,train,inference}.py`
Configuration	`English/config.yaml`	`Telugu/config.yaml`
Checkpoint (weights only, Git LFS)	`English/checkpoints/model_only_20000.pth.tar`	`Telugu/checkpoints/model_only_25000.pth.tar`
Loss plots	`Results/English/{training,validation}_loss.png`	`Results/Telugu/{training,validation}_loss.png`
Test audio	`Results/English/test_generated_audios/`	`Results/Telugu/test_generated_audios/`
Variance plots	`Results/English/variance_adaptor_plots/{duration,pitch,energy}/`	`Results/Telugu/variance_adaptor_plots/{duration,pitch,energy}/`
Loss logs, plotted values, evaluation results	`English/logs/`	`Telugu/logs/`
Split manifests, feature index, aligned pronunciations	`English/data/`	`Telugu/data/`

The reference implementation (`model/`, `audio/`, `text/`, `transformer/`, `utils/`, `preprocessor/`, `hifigan/`, `lexicon/`) stays at the repository root, because both language folders import it.

7.2 Software

- **Python environment:** Python 3.10 with PyTorch 2.11.0. The other dependencies are pinned in `requirements.txt`: NumPy 2.2.6, librosa 0.10.2.post1, pyworld 0.3.5, tgt 1.5, g2p-en 2.1.0, soxr 1.1.0, Hugging Face `datasets` 5.1.0, and others.
- **Forced aligner:** Montreal Forced Aligner 3.4.3, in a separate conda-forge environment (`README.md`).
- **Speech recognition:** only the evaluation scripts need it: `pip install openai-whisper`, model `small`.
- **Seed:** 42 for data selection, training and evaluation sampling.

7.3 Commands

Run from the repository root.

Inference with the submitted checkpoints (the voice and pitch range come from the configs):

```
python English/inference.py --checkpoint English/checkpoints/model_only_20000.pth.tar --test_sentences
python Telugu/inference.py --checkpoint Telugu/checkpoints/model_only_25000.pth.tar --test_sentences
python English/inference.py --checkpoint English/checkpoints/model_only_20000.pth.tar --text "Your
sentence." --id mine
python Telugu/inference.py --checkpoint Telugu/checkpoints/model_only_25000.pth.tar --text "మీ వాక్యం." --
id mine
```

English data preparation, training and voice choice:

```
python English/select_clean_split.py candidates
python English/preprocess.py export
python English/preprocess.py dict
python English/preprocess.py align --jobs 4
python English/select_clean_split.py select
python English/preprocess.py features
python English/preprocess.py pronunciations
python English/make_group_variant.py
python English/train.py --require_gpu
python English/plot_losses.py
python English/export_checkpoint.py --step 20000
python English/rank_voices.py --candidates 10 --sentences 30
```

Telugu data preparation and training:

```
python Telugu/inspect_dataset.py
python Telugu/preprocess.py split
python Telugu/preprocess.py export
python Telugu/preprocess.py dict
python Telugu/preprocess.py align --jobs 4
python Telugu/preprocess.py features
python Telugu/train.py --require_gpu --total_step 25000
python English/plot_losses.py --config Telugu/config.yaml
python English/export_checkpoint.py --config Telugu/config.yaml --step 25000
```

Evaluation (both languages; `--config Telugu/config.yaml` for Telugu):

```
python English/evaluate_asr.py
python English/evaluate_mcd.py
```

Tests (20, all passing):

```
python -m unittest discover -s English/tests
python -m unittest discover -s Telugu/tests
```